13. Which of the following is acceptor impurity element?

(a) Antimony (b) Gallium (c) Arsenic (d) Phosphorous

14. In N-type semiconductor free electrons are the ……. Carriers

(a) Minority (b) Majority (c) Magnetic (d) Neutral

15. A doped semiconductor is called

(a) impure semiconductor (b) intrinsic semiconductor

(c) Pure semiconductor (d) Extrinsic semiconductor

II. Answer in one or few Sentences

1. What is an atomic number?
2. What is atomic weight?
3. What is valence electron?
4. Draw the atomic structure of germanium atom.
5. What is called energy band?
6. What is meant by electron emission?
7. Write active components with examples.
8. What is meant by doping?

III Explain the following questions in one or two paragraph. (3 marks)

1. Write short notes on free electrons.
2. Explain energy bands.
3. Explain conductor, semiconductor and insulator.
4. Explain formation of n-type semiconductor with diagram.
5. Explain formation of p-type semiconductor with diagram.
6. Explain the mechanism of current flow in a forward biased pn junction.

IV Describe the following questions in a page. (10 marks)

1. Explain the Bohr's atomic model with neat diagram.
2. Explain the different types of electron emission with neat diagrams.
3. Explain the forward biasing of pn junction with necessary diagram.
4. Explain the reverse biasing of pn junction with necessary diagram.

Answers

1) **c**	2) **b**	3) **c**	4) **a**	5) **b**
6) **d**	7) **c**	8) **a**	9) **c**	10) **d**
11) **a**	12) **c**	13) **b**	14) **b**	15) **d**

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